

Unit 101 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 97.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 45 plots, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (SE 1.9 %)	Type	Sierran mixed conifer
Trees per acre	123	CWHR size / density	4 / D (cover 83 %)
QMD	17.8 in	SDI	288 (51 % of max 567)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	203 sq ft/ac 19.8 in	BA QMD	9 sq ft/ac 7.6 in
TPA	95	TPA	28
Volume	39.7 MBF/ac (7212 cu ft)	Volume	4.2 green tons/ac (140 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~146 sq ft/ac BA	Sampling error	3.9 % at 95 % (t = 2.02, n = 45)
Species (BA)	PP 70, WF 60, IC 32, DF 29, SP 21	Stratum standard	40 % - MEETS; 45 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	8	15.6	3.6	47	0.3
8-11.9	34	27.7	15.1	290	1.6
12-15.9	74	30.7	32.9	801	4.4
16-19.9	82	21.2	36.4	1074	5.9
20-23.9	68	11.7	30.2	1015	5.6
24-27.9	50	6.0	22.2	814	4.5
28-31.9	52	4.8	23.1	945	5.2
32-35.9	33	2.3	14.7	632	3.5
36-39.9	25	1.4	11.1	527	2.9
40-43.9	18	0.9	8.0	384	2.1
44-47.9	10	0.4	4.4	226	1.2
48-51.9	3	0.1	1.3	73	0.4
52-55.9	9	0.3	4.0	227	1.2
56-59.9	4	0.1	1.8	112	0.6
60-63.9	7	0.2	3.1	185	1.0
All	477	123.4	212.4	7352	40.4

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	158	45.2	70.2	16.9	2404	13.2
WF	134	31.5	59.6	18.6	2061	11.3
IC	71	19.1	31.6	17.4	1097	6.0
DF	66	14.3	29.3	19.4	1077	5.9
SP	47	13.3	20.9	17.0	689	3.8
PPP	1	0.1	0.4	35.1	23	0.1
All	477	123.4	212.4	17.8	7352	40.4

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101: plots on the 300 ft grid (orange = QA flag)

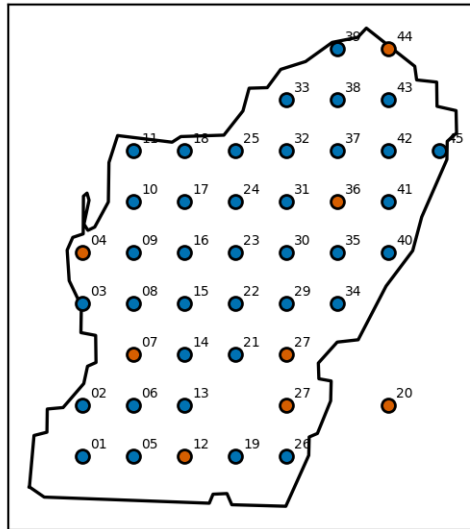


Figure 1. Plot locations with QA status.

QA findings: 7 item(s) need crew follow-up

Plot	Finding
101-04	tree 1: unknown species code 'PPP'
101-07	tree 9: height 32 ft implausible for DBH 27.8
101-12	tree 1: DBH 999.0 out of range
101-20	plot location falls outside its unit boundary (GPS or unit number error)
101-27	duplicate plot id recorded 2 times
101-36	tree 2: height missing
101-44	tree 1: height 15 ft implausible for DBH 34.0

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
101-01	12	240	49	ok
101-02	10	200	63	ok
101-03	6	120	112	ok
101-04	9	180	133	FLAG
101-05	9	180	59	ok
101-06	12	240	60	ok
101-07	13	260	16	FLAG
101-08	11	220	79	ok
101-09	11	220	31	ok
101-10	12	240	107	ok
101-11	9	180	125	ok
101-12	10	200	23	FLAG
101-13	11	220	23	ok
101-14	10	200	55	ok
101-15	10	200	18	ok
101-16	11	220	20	ok
101-17	11	220	15	ok
101-18	10	200	22	ok
101-19	10	200	45	ok
101-20	11	220	111	FLAG
101-21	11	220	4	ok
101-22	10	200	16	ok
101-23	11	220	32	ok
101-24	11	220	53	ok
101-25	11	220	17	ok
101-26	15	300	68	ok
101-27	11	220	44	FLAG
101-27	9	180	60	FLAG
101-29	12	240	26	ok
101-30	11	220	17	ok
101-31	11	220	46	ok
101-32	11	220	18	ok
101-33	9	180	57	ok
101-34	9	180	27	ok
101-35	10	200	27	ok
101-36	12	240	26	FLAG
101-37	11	220	20	ok
101-38	11	220	44	ok
101-39	10	200	44	ok
101-40	12	240	13	ok
101-41	12	240	15	ok
101-42	10	200	12	ok
101-43	9	180	46	ok
101-44	11	220	20	FLAG
101-45	10	200	52	ok